



Research Paper

Research on adaptive compensation methods for infrared temperature sensing errors in drones for wildfire monitoring

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ABSTRACT

Airborne infrared thermography is promising for wildfire monitoring, but field measurements are often degraded by coupled disturbances such as smoke occlusion, time-varying observation geometry, and platform perturbations. These effects introduce both systematic temperature bias and short-term abnormal fluctuations, reducing the reliability of infrared temperature sequences for subsequent fire analysis. This paper proposes a two-step framework combining multi-factor joint compensation and combustion-stage-adaptive stabilization. First, observation range, viewing angle, and smoke transmittance are jointly incorporated into an interpretable compensation model to correct systematic drift. Second, a stage-label-driven stabilization strategy is introduced to suppress residual anomalies and high-frequency perturbations by adaptively adjusting outlier suppression and exponential smoothing strength. Real-fire experiments with a UAV-mounted multi-sensor payload and thermocouple references show that the proposed framework improves agreement with the reference sequence in both full-process and stage-wise evaluations. In the full-sequence assessment, the bias is reduced from $-49.11\text{ }^{\circ}\text{C}$ to $3.94\text{ }^{\circ}\text{C}$, while the correlation increases to 0.949 after stabilization. Comparison with polynomial compensation models further shows that the proposed method outperforms conventional compensation-oriented models. Additional validation under different real-fire conditions confirms that the same two-step improvement pattern is retained in a second fire scenario with clearly different field and combustion characteristics.

1. Introduction

Forest and grassland wildfires spread rapidly and evolve under complex, heterogeneous conditions. Accurate fire-scene temperature information is therefore of great importance for assessing combustion dynamics and supporting suppression decision-making [1]. In recent years, the integration of unmanned aerial vehicle (UAV) platforms with thermal infrared (TIR) cameras has provided an effective technical route for large-area, non-contact wildfire temperature monitoring [2]. Related studies have also shown the value of UAV systems in wildfire management [3] and the practical utility of UAS-derived infrared imagery for evaluating fire behavior and effects [4].

Despite these advances, accurate TIR temperature retrieval in real wildfire scenes remains challenging. In continuous aerial observation, the measurement process is often affected by coupled disturbances, including time-varying observation distance, off-axis viewing geometry, spatially non-uniform smoke occlusion, and platform-induced

perturbations. Valero et al. [5] addressed the issue of thermal infrared video stabilization, showing that motion-related artifacts significantly affect airborne thermal observation. Moreover, the apparent temperature from a TIR camera can deviate from the true target temperature due to variations in radiative transfer, target emissivity, and environmental radiance, as highlighted in classical radiation thermometry models [6,7].

A substantial body of work has examined error mechanisms and correction strategies in infrared thermometry. Classic treatments and practical overviews provide the theoretical and methodological basis for calibration and uncertainty analysis in radiometric temperature measurement [8,9]. Beyond general calibration, recent studies have emphasized that measurement errors are strongly scene- and configuration-dependent: incomplete target coverage within the field of view can introduce systematic bias and requires dedicated compensation [10], while environmental media such as fog or smoke further modulate the received radiation through attenuation and path

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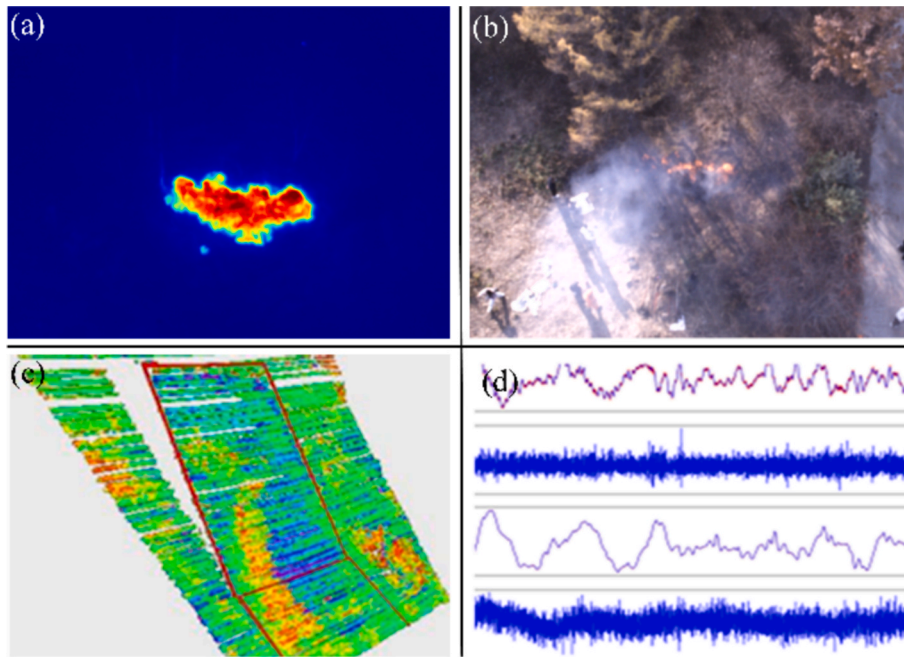


Fig. 1. Data samples: (a) Thermal infrared image; (b) visible light image; (c) point cloud; (d) IMU data visualization.

contributions, leading to additional temperature deviations [11,12]. Meanwhile, observation geometry is another key driver of bias, and angle-related effects can be mitigated when auxiliary geometric information is incorporated into the correction procedure [13]. For more physically complete modeling of the propagation path, atmospheric scattering terms have also been quantified by combining active ranging with radiative-transfer simulation [14].

Integrated strategies have increasingly been explored to address multi-factor coupling under complex interferences. Pan et al. proposed an intelligent compensation method for infrared temperature measurement under multiple interference factors, showing that realistic measurement errors are often difficult to handle using only single-factor correction [15]. Pan et al. further reviewed the major interference factors and compensation methods in infrared thermography, and emphasized that robust correction in complex environments generally requires joint consideration of multiple disturbance sources [16]. In addition, Pan et al. developed a transmittance-correction-based method under dynamic water-mist interference, indicating that visible-light observations can provide useful medium-state information for infrared temperature compensation [17]. Along a similar line, calibration-oriented studies have improved infrared measurement accuracy through pixel-wise in situ calibration [18], high-angle-view calibration [19], and data-driven calibration models [20].

Recent cross-modality reconstruction studies also provide useful methodological inspiration. Enhanced deep image priors [21], cross-modality mutual learning with interactive autoencoders [22], and model-guided coarse-to-fine fusion strategies [23] all suggest that when the target quantity cannot be reliably recovered from a single degraded sensing channel, cross-modality information and model-guided constraints can improve robustness. For the present study, these works suggest that cross-modality information and model-guided constraints can improve robustness when medium degradation cannot be reliably characterized from a single sensing channel.

Even after reducing systematic biases, wildfire infrared temperature sequences still exhibit abnormal fluctuations. Such fluctuations may compromise data interpretability and downstream application value, particularly in combustion characteristic analysis and fireline research. Signal-processing techniques have long been applied in thermal imaging to enhance detectability and reduce noise interference. Chung et al. [24]

reviewed signal-processing advancements for pulsed thermography, while Balageas et al. [25] introduced the thermographic signal reconstruction (TSR) method, which enhances signal quality while preserving trends. In addition, wildfire combustion is typically stage-dependent: Santoso et al. [26] examined the transition from smouldering to flaming combustion, and Wang et al. [27] studied combustion-phase detection using remote sensing. These studies indicate that combustion-stage information can provide a meaningful temporal label, which can be leveraged to adapt stabilization strength according to different combustion phases.

Motivated by these observations, this paper focuses on two issues that are not jointly handled well by existing methods in UAV-based wildfire thermometry: multi-factor systematic bias and combustion-stage-dependent transient volatility. Existing compensation-oriented methods mainly address bias correction under one or several disturbance factors, whereas temporal processing studies mainly support sequence enhancement or stage interpretation. In contrast, the present work is designed to address both aspects within a unified framework. Its main methodological differences are twofold: first, it performs joint compensation for distance, viewing-angle, and smoke-related degradation rather than treating only a single disturbance source; second, it links systematic compensation with stage-aware temporal stabilization instead of handling bias correction and fluctuation suppression as two separate procedures.

To address these issues, this paper proposes a two-step adaptive framework for infrared temperature-sensing error compensation in UAV-based wildfire monitoring. Specifically, a radiance-increment-based joint compensation model is first constructed to reduce systematic bias caused by observation distance, viewing geometry, and smoke-related degradation. A combustion-stage-adaptive stabilization strategy is then introduced to suppress residual short-term anomalies while preserving the underlying thermal evolution trend. This design aims to improve both temperature-estimation consistency and sequence reliability in dynamic UAV-based wildfire observations.

The remainder of this paper is organized as follows. Section 2 presents the methodology and implementation details. Section 3 introduces the experimental design and evaluation protocol. Section 4 presents the results and discussion.

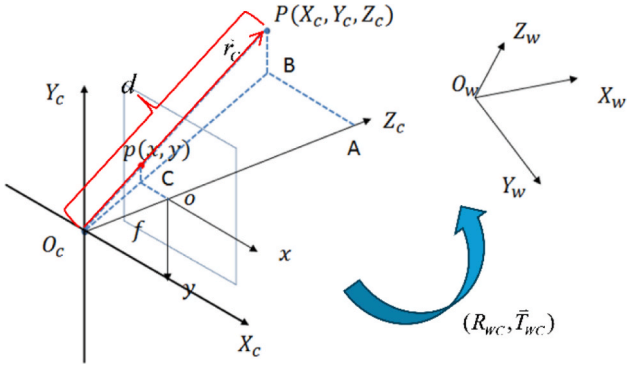


Fig. 2. Schematic diagram of observation geometry parameters: The parameters d and r_c are indicated by red lines.

2. Methodology

2.1. Quantification of influencing factors

A UAV-borne multi-sensor system is used to synchronously acquire thermal infrared (TIR) imagery, co-view visible imagery, LiDAR point clouds, and navigation states. Representative multi-modal samples are shown in Fig. 1. In this work, each TIR pixel and its associated 3D point are treated as the basic processing unit, enabling pixel-wise alignment between the temperature field and the key influencing factors.

To support the subsequent joint compensation, three key factors are quantified at the pixel level: observation distance d , viewing angle θ , and smoke transmittance τ_s . The definitions and acquisition procedures are described below.

2.1.1. Observation geometry: Distance and viewing angle

Let $\{W\}$, $\{L\}$, and $\{C\}$ denote the world, LiDAR, and TIR camera coordinate frames, respectively. The upward unit vector of $\{W\}$ is $\mathbf{e}_z = [0, 0, 1]^T$. A fixed rigid transformation from $\{L\}$ to $\{C\}$ is obtained via extrinsic calibration and represented by $\mathbf{R}_{CL} \in \mathbb{R}^{3 \times 3}$ and $\mathbf{t}_{CL} \in \mathbb{R}^3$.

For a 3D point $\mathbf{p}_L = [X_L, Y_L, Z_L]^T$ expressed in $\{L\}$, the observation distance is defined as the Euclidean norm of the corresponding point in $\{C\}$:

$$d = \|\mathbf{R}_{CL}\mathbf{p}_L + \mathbf{t}_{CL}\|_2 \quad (1)$$

For a TIR pixel coordinate (u, v) , the normalized line-of-sight direction in $\{C\}$ is computed using the camera intrinsic matrix $\mathbf{K}$:

$$\mathbf{r}_c = \frac{\mathbf{K}^{-1}[u, v, 1]^T}{\|\mathbf{K}^{-1}[u, v, 1]^T\|_2} \quad (2)$$

With the navigation-provided camera attitude and mounting extrinsic, the rotation from $\{C\}$ to $\{W\}$ is denoted by $\mathbf{R}_{WC}$, and the line-of-sight direction in $\{W\}$ is

$$\mathbf{r}_w = \mathbf{R}_{WC}\mathbf{r}_c \quad (3)$$

Accordingly, the viewing angle is defined as the angle between $\mathbf{r}_w$ and the reference normal direction $\mathbf{e}_z$:

$$\theta(u, v, t) = \arccos\left(\frac{\mathbf{r}_w \cdot \mathbf{e}_z}{\|\mathbf{r}_w\|_2}\right) \quad (4)$$

The definitions of d and θ , together with the coordinated transformations involved, are illustrated in Fig. 2.

2.1.2. Pixel-wise smoke transmittance estimation

In wildfire environments, smoke absorption and scattering attenuate the transmitted target radiation and introduce path radiance, which can bias TIR temperature measurements. Since it is difficult to robustly isolate smoke-induced attenuation using only the TIR temperature

sequence, co-view visible imagery is introduced to indirectly characterize the degree of smoke occlusion along the line of sight [17]. Following the general idea of using visible observations to provide medium-state information for infrared temperature compensation, a pixel-wise smoke descriptor is constructed and used as the medium-related input of the joint model.

To reduce the influence of irrelevant background variations, smoke estimation is performed only within the fire-related region. Specifically, a fire-related region of interest (ROI) is first extracted from the TIR image and then mapped to the visible image by camera extrinsic calibration. Let I_{ref} denote the reference brightness under smoke-free or weak-smoke conditions, and let $I(t)$ denote the brightness at time t . A normalized smoke index is defined as

$$s(t) = \max\left(0, \frac{I_{\text{ref}} - I(t)}{I_{\text{ref}} + \varepsilon}\right) \quad (5)$$

where $\varepsilon > 0$ is a small constant to avoid an excessively small denominator. This definition ensures that $s(t)$ increases monotonically as visible brightness decreases under stronger smoke occlusion, while the non-negativity constraint excludes physically implausible negative occlusion.

Based on the Beer–Lambert attenuation form, the smoke index is mapped to the pixel-wise smoke transmittance:

$$\tau_s(t) = \exp(-s(t)) \quad (6)$$

which guarantees $\tau_s \in (0, 1]$ and enforces a monotonic decrease with increasing smoke occlusion. It should be noted that τ_s here is not directly interpreted as the true physical transmittance in the thermal infrared band. Instead, it is a visible-derived smoke descriptor that characterizes the relative strength of medium occlusion along the line of sight. The correspondence between this visible-band descriptor and thermal-band attenuation is established later through a calibrated effective mapping in the joint compensation model, rather than by directly assuming spectral equivalence.

Finally, τ_s is resampled to the TIR image coordinates to obtain a pixel-wise medium descriptor aligned with the temperature field. In implementation, light spatial smoothing can be applied to suppress local flicker and sensor noise.

2.2. Multi-factor joint compensation model and parameter identification

In dynamic wildfire observations, the apparent TIR temperature is jointly affected by observation distance, viewing geometry, and smoke along the line of sight. To describe these coupled effects in a compact and interpretable way, we work in an equivalent radiance-increment domain relative to the ambient background and model the measurement as a target increment scaled by an equivalent transfer coefficient.

2.2.1. Equivalent radiance-increment formulation

Within a narrow spectral band and a limited temperature range, the effective radiative flux received by an infrared imager can be approximated by a generalized power-law form:

$$\Phi(T) \propto T^n \quad (7)$$

where $\Phi(T)$ denotes the effective radiative flux corresponding to temperature T , and n is an effective exponent. In this work, all temperatures are converted to Kelvin and $n = 4$ is adopted.

Let $T_{\text{IR}}(u, v, t)$ be the TIR-measured temperature at pixel (u, v) and time t , and let $T_{\text{env}}(t)$ be the ambient of background equivalent temperature. Using the ambient background as reference, the radiative-flux increment along the viewing direction can be written as

$$\Delta\Phi_{\text{IR}}(u, v, t) = \Phi(T_{\text{IR}}(u, v, t)) - \Phi(T_{\text{env}}(t)) \propto T_{\text{IR}}^n(u, v, t) - T_{\text{env}}^n(t) \quad (8)$$

Based on an equivalent one-dimensional transfer representation, the

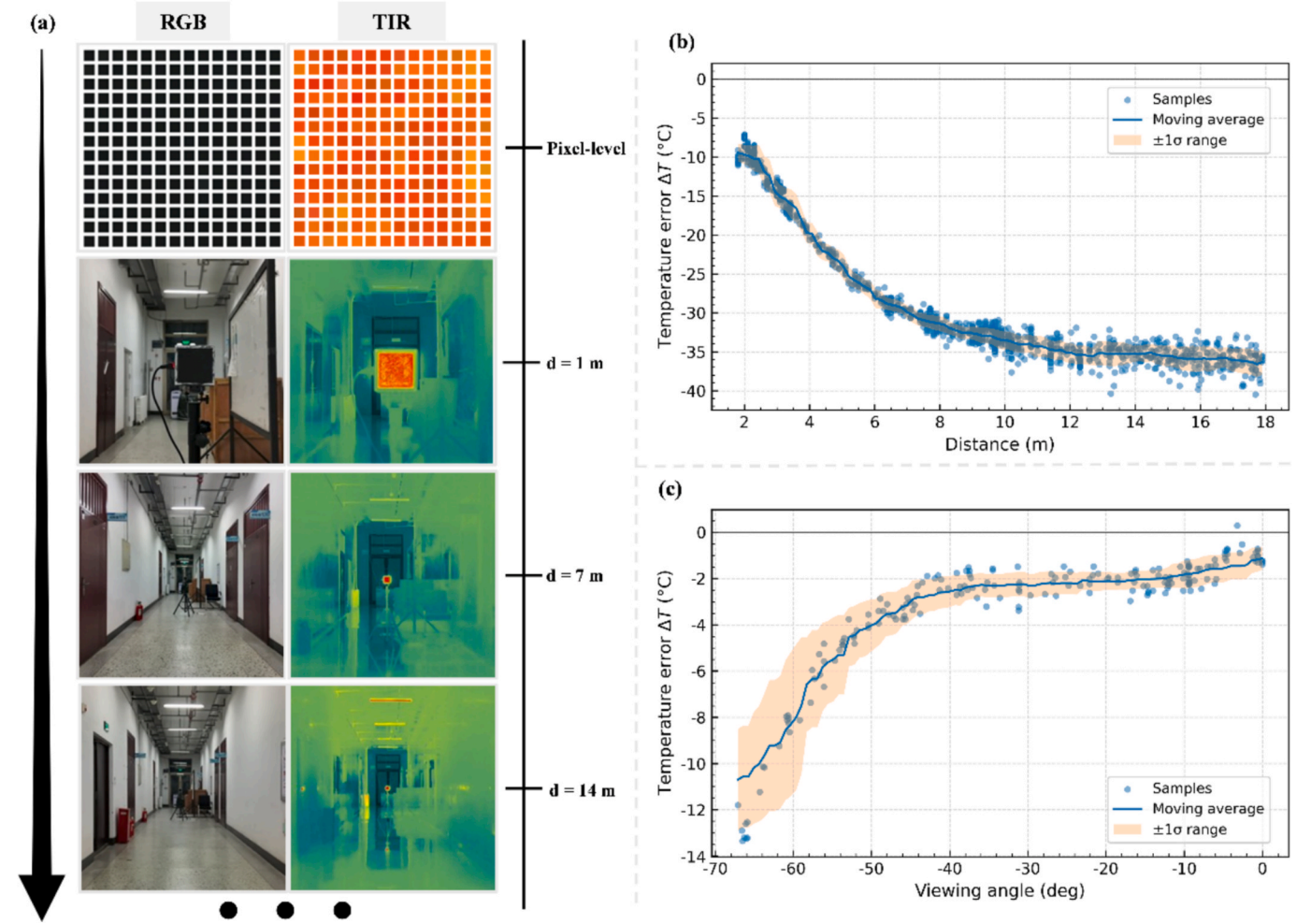


Fig. 3. Geometric calibration data and error distribution under indoor smoke-free, constant-temperature conditions: (a) Representative setup for indoor smoke-free, constant-temperature calibration (the field-of-view calibration uses the same apparatus and target, so its setup diagram is not shown separately); (b) scatter plot of temperature error versus distance; (c) scatter plot of temperature error versus field of view.

increment reaching the detector is modeled as the target increment scaled by an equivalent transfer coefficient $\eta(u, v, t) \in (0, 1]$:

$$T_{\text{IR}}^n(u, v, t) - T_{\text{env}}^n(t) = \eta(u, v, t) [T_{\text{obj}}^n(u, v, t) - T_{\text{env}}^n(t)] \quad (9)$$

where $T_{\text{obj}}(u, v, t)$ is the target-equivalent temperature to be recovered.

Strictly speaking, the equivalent transfer coefficient in Eq. (9) should be regarded as a joint function $\eta = \eta(d, \theta, \tau_s)$, reflecting the combined influence of observation distance, viewing angle, and smoke-related attenuation. In the present work, Eq. (10) is not treated as an exact analytical factorization of the full radiative-transfer process. Instead, it is introduced as a first-order separable approximation in the radiance-increment domain around a reference operating condition. Specifically, letting $g(d, \theta, \tau_s) = \ln \eta(d, \theta, \tau_s)$, a local first-order approximation is adopted and only the main-effect terms of d , θ , and τ_s are retained, while higher-order interaction terms are absorbed into the residual modeling error. Exponentiation of this approximation leads to the following product form:

$$\eta(u, v, t) \approx \eta_d(d(u, v, t)) \eta_\theta(\theta(u, v, t)) \eta_s(\tau_s(u, v, t)) \quad (10)$$

Therefore, Eq. (10) should be understood as a compact first-order approximation that preserves the dominant effects, rather than as a direct or exact physical factorization.

Combining (9)–(10), the joint-compensated temperature is obtained by solving for T_{obj} :

$$T_{\text{comp}}(u, v, t) = \left[\frac{T_{\text{IR}}^n(u, v, t) - T_{\text{env}}^n(t)}{\eta_d(d) \eta_\theta(\theta) \eta_s(\tau_s)} + T_{\text{env}}^n(t) \right]^{1/n} \quad (11)$$

where $\eta_d(d)$, $\eta_\theta(\theta)$, and $\eta_s(\tau_s)$ are evaluated at (u, v, t) for brevity. Equation (11) provides a pixel-wise compensation that explicitly integrates distance, viewing geometry, and smoke attenuation.

2.2.2. Parameterization and identification of sub-transfer functions

The geometric terms capture radiance weakening caused by distance-dependent extinction and viewing-geometry effects. A calibration experiment under smoke-free conditions with a constant-temperature target is conducted to identify these terms. For calibration samples, a transfer-coefficient observation is constructed in the radiance-increment domain. Let $T_{\text{ref}}(t)$ denote the reference target temperature. Then

$$\hat{\eta}(t) = \frac{T_{\text{IR}}^n(t) - T_{\text{env}}^n(t)}{T_{\text{ref}}^n(t) - T_{\text{env}}^n(t)} \quad (12)$$

By varying d while keeping θ approximately constant, the variation of $\hat{\eta}$ mainly reflects the distance-related effect; by varying θ while keeping d approximately constant, the variation of $\hat{\eta}$ mainly reflects the viewing-angle-related effect. The calibration results and the fitted trends are shown in Fig. 3, where stable monotonic attenuation behavior can be observed in these two controlled settings.

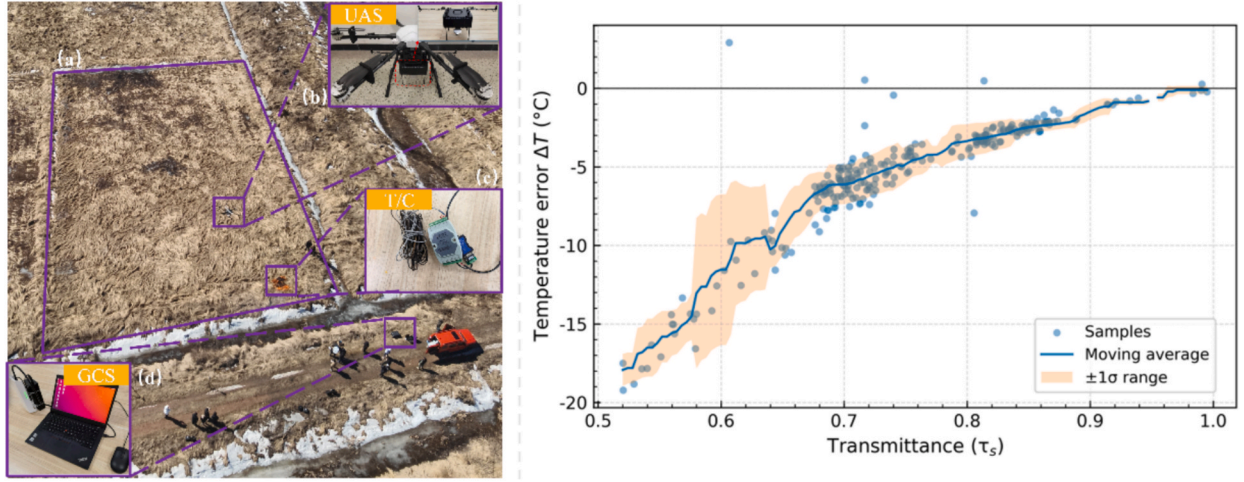


Fig. 4. Relationship between the smoke dataset and temperature errors: the left half depicts the collection scenario for the real wildfire smoke dataset, where (a) the overall test environment, (b) the unmanned aerial system (UAS), (c) the thermocouple (T/C) deployment location, and (d) the ground control station (GCS) are shown; the right half shows a scatter plot of temperature error versus smoke transmittance.

Table 1

Fitted parameters of the sub-transfer functions in the joint model.

Child item	Parameters	Data Source	Estimated value	Confidence interval	Scope of Application
$\eta_d(d)$	k_d	Indoor constant temperature, smoke-free	0.0561	(0.055, 0.057)	$d \in (0, 18)$
$\eta_\theta(\theta)$	k_θ	Indoor constant temperature, smoke-free	0.193	(0.158, 0.228)	$\theta \in (-70, 70)$
$\eta_s(\tau_s)$	k_s	Real grassland fire scene	0.396	(0.325, 0.467)	Mild-Medium

Based on these observed trends, the following pragmatic parameterization is adopted:

$$\eta_d(d) = \exp(-k_d(d - d_0)), \eta_\theta(\theta) = \cos(\theta)^{k_\theta} \quad (13)$$

where d_0 is a reference distance, k_d is the distance extinction parameter, and k_θ is the viewing-angle attenuation exponent. The parameters k_d and k_θ are identified by least-squares fitting of (13) to $\hat{\eta}(t)$.

For smoke-involved samples, the overall transfer coefficient $\hat{\eta}(t)$ is first computed by (13). With the calibrated geometric terms, the smoke-dominant component is isolated as

$$\hat{\eta}_s(t) = \frac{\hat{\eta}(t)}{\eta_d(d(t))\eta_\theta(\theta(t))} \quad (14)$$

where $\hat{\eta}_s(t)$ represents the equivalent attenuation component after removing the distance-related and viewing-angle-related effects. The paired samples $\{\hat{\eta}_s(t), \tau_s(t)\}$ are shown in Fig. 4, where a stable monotonic relationship can be observed after excluding obvious outliers caused by abrupt occlusion or unstable measurements.

According to this observed trend, the smoke term is modeled by a monotonic, bounded one-parameter function:

$$\eta_s(\tau_s) = \tau_s^{k_s} \quad (15)$$

where k_s is the effective smoke attenuation exponent. The parameter k_s is estimated by least squares using the paired samples $\{\hat{\eta}_s(t), \tau_s(t)\}$. In this formulation, Eq. (15) provides a calibrated mapping from the visible-derived smoke descriptor to an equivalent thermal attenuation coefficient.

With the identified parameters, the joint-compensated temperature $T_{\text{comp}}(u, v, t)$ is computed by (11) for field experiments. The identified parameters used in the joint transfer model are summarized in Table 1.

2.3. Combustion-stage-aware stabilization

After the joint compensation in Section 2.2, the systematic bias

induced by distance, viewing geometry, and smoke attenuation is substantially corrected. However, in real wildfire observations, transient smoke occlusion, abrupt flame-shape changes, and platform micro-perturbations may still introduce short-term abnormal values and high-frequency disturbances. Such disturbances are weakly related to the true combustion trend; if not suppressed, they can degrade analysis accuracy and weaken the ability of the IR sequence to follow the main trend of the thermocouple reference. Therefore, we further introduce a combustion-stage-aware stabilization module, where a combustion-stage label $S(t)$ controls the strength of outlier suppression and smoothing.

Let $T_{\text{comp}}(t)$ denote the Comp (joint-compensated) temperature sequence at a given aligned pixel, and let $T_{\text{supp}}(t)$ denote the stabilized output. The proposed stabilization consists of two cascaded steps: (i) robust outlier suppression based on a Hampel filter with MAD-based scale estimation, producing an intermediate sequence $\tilde{T}(t)$; and (ii) exponentially weighted moving average (EWMA) applied to $\tilde{T}(t)$ to obtain $T_{\text{supp}}(t)$. Key parameters in both steps are switched according to the stage label $S(t)$.

2.3.1. Stage labeling of the combustion process

The wildfire combustion process is divided into three stages, namely the preignition combustion phase, flaming combustion phase, and smoldering combustion phase. These stages mainly differ in (i) the temperature level and (ii) the intensity of short-term fluctuations in continuous observations. We define a statistics-based stage labeling method using two features: normalized temperature level and a local robust fluctuation index.

Given a temperature sequence $T(t)$ from the same run, we use in practice the compensated IR sequence $T_{\text{comp}}(t)$ to avoid relying on thermocouple information at deployment, and we first compute the normalized temperature level:

$$T_n(t) = \frac{T(t) - \min(T)}{\max(T) - \min(T) + \epsilon} \quad (16)$$

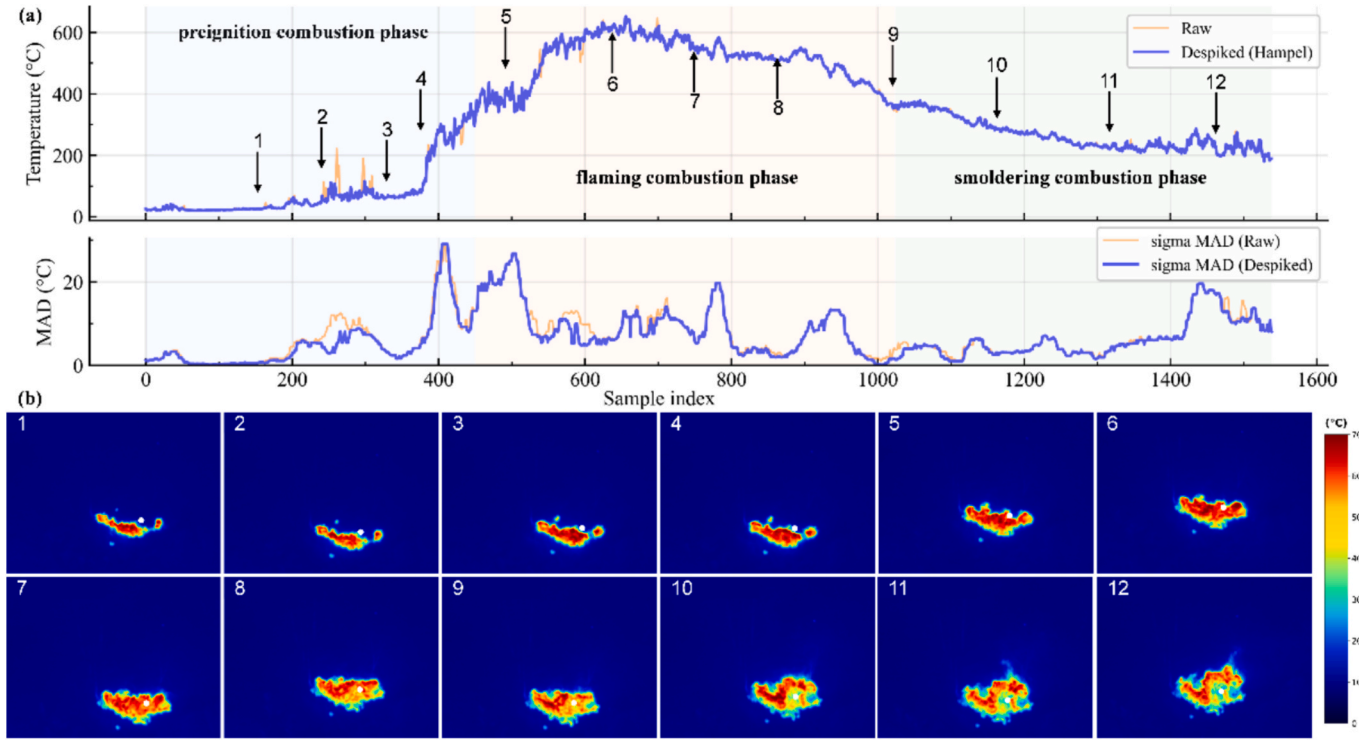


Fig. 5. Continuous temperature sequence observations at the fire scene: (a) Temperature sequence and corresponding fluctuation intensity at each time point; (b) twelve infrared thermal images sequentially selected from the combustion process, corresponding to the time points indicated by numbered arrows in (a). Observation locations are marked with white dots to correlate with the combustion stage at each observation point.

where $\varepsilon > 0$ is a small constant preventing an excessively small denominator.

To characterize short-time-scale fluctuation intensity, a sliding window $W(t)$ of approximately 1s in length is used to compute a local robust fluctuation index:

$$\sigma(t) = \text{MAD}(T(W(t))) \quad (17)$$

where $\text{MAD}(\cdot)$ denotes the median absolute deviation relative to the median of the window.

Thresholds are determined by fixed-quantile rules for each run. Let $Q_p(\cdot)$ denote the p -quantile operator. The thresholds are defined as

$$T_L = Q_{0.25}(T_n(t)), T_H = Q_{0.75}(T_n(t)), \sigma_H = Q_{0.65}(\sigma(t)), \sigma_L = Q_{0.25}(\sigma(t)) \quad (18)$$

Based on these thresholds, stage candidate criteria are defined as:

- (1) Preignition combustion phase candidate: $T_n(t) \leq T_L$ and $\sigma(t) \leq \sigma_L$.
- (2) Flaming combustion phase candidate: $T_n(t) \geq T_H$ or $\sigma(t) \geq \sigma_H$.
- (3) Smoldering combustion phase candidate: not satisfying the flaming candidate, with temperature falling back and becoming stable; constrained by $T_n(t) \leq T_H$ and $\sigma(t) \leq \sigma_H$.

To reflect the stage continuity of combustion and to avoid instantaneous misclassification, a minimum consecutive-sample constraint is applied: the stage label is updated only when the corresponding candidate criterion holds for N_c consecutive samples. The final stage label is denoted as $S(t)$, which serves as the control signal for the stage-adaptive stabilization described next. The continuous temperature sequence and the resulting stage segmentation are illustrated in Fig. 5.

2.3.2. Robust outlier suppression

Robust outlier suppression follows the Hampel-filter principle and uses stage-dependent parameters to control the suppression strength.

Table 2

Parameters for Staged Stabilization.

$S(t)$	w_H	k_H	α
Preignition combustion phase	7	3.0	0.25
Flaming combustion phase	5	2.5	0.45
Smoldering combustion phase	7	2.5	0.35

Specifically, a stage-dependent window half-length $w_H(S(t))$ and a stage-dependent threshold coefficient $k_H(S(t))$ are used. For the current time t , define the window $\Omega_t = [t - w_H, t + w_H]$. Let m_t be the median of $T_{\text{comp}}(\tau)$ within Ω_t , and let MAD_t be the corresponding median absolute deviation. A robust scale estimate is computed as $\hat{\sigma}_t = 1.4826\text{MAD}_t + \varepsilon$. The outlier decision and replacement are given by

$$\tilde{T}(t) = \begin{cases} m_t, & |T_{\text{comp}}(t) - m_t| > k_H(S(t))\hat{\sigma}_t \\ T_{\text{comp}}(t), & \text{otherwise} \end{cases} \quad (19)$$

where $\tilde{T}(t)$ is the intermediate sequence after robust suppression.

2.3.3. Trend-preserving smoothing

To further suppress residual high-frequency disturbance, EWMA is applied to $\tilde{T}(t)$. The smoothing factor $\alpha(S(t)) \in (0, 1)$ is switched according to the stage label:

$$T_{\text{supp}}(t) = \alpha(S(t))\tilde{T}(t) + [1 - \alpha(S(t))]T_{\text{supp}}(t-1), t \geq 2 \quad (20)$$

with initialization $T_{\text{supp}}(1) = \tilde{T}(1)$ s. To guarantee output continuity at stage transitions, EWMA uses a single-state recursion: only α is updated when switching stages, while the recursion state remains continuous, avoiding boundary jumps.

A reproducible set of stage-wise parameters validated on real wild-fire data is summarized in Table 2.



Fig. 6. Real-fire experiment layout and observation geometry: (a) The overall test environment; (b) the ground control station (GCS); (c) the unmanned aerial system (UAS); (d) the thermocouple (T/C) deployment.

3. Experiments and results

3.1. Real-fire experiment and data acquisition

A real-fire burning experiment was conducted to validate the proposed method under practical off-axis observation conditions. A UAV-borne multi-sensor payload was used to continuously record thermal infrared (TIR) image sequences together with synchronized co-view visible images, LiDAR point clouds, and navigation states. Meanwhile, K-type thermocouples were deployed within the area influenced by the fire front to provide contact reference temperature sequences.

Because thermocouples provide point-wise contact measurements while the TIR camera reports an apparent temperature field along the line of sight, a dedicated spatial-temporal alignment procedure was designed to ensure traceable and consistent comparisons. Before ignition, the test site was reconstructed in 3D using the onboard LiDAR and navigation system. For each thermocouple, its location was measured using both a ground RTK pole and the UAV system and then cross-checked to reduce systematic positioning errors. The experimental layout and observation geometry are shown in Fig. 6.

During burning, thermocouples are often difficult to directly identify in the visible and TIR frames due to flames and smoke. Therefore, their pre-measured 3D coordinates were used together with the real-time UAV pose and the calibrated camera-LiDAR extrinsic to back-project each thermocouple position onto the TIR imaging plane, yielding the corresponding pixel location over time. To improve robustness against pixel-level noise and minor misalignment, the TIR temperature used for comparison was computed as the spatial average over a 3×3 neighborhood centered at the projected pixel.

All sensor streams were synchronized under a unified onboard time base, and the analysis was performed on the aligned timestamps to obtain comparable one-dimensional temperature sequences for subsequent evaluation.

3.2. Validation protocol and evaluation metrics

The thermocouple measurement is used as the reference sequence, denoted as $T_{\text{ref}}(t)$. The TIR-derived comparison sequence at the same timestamps is evaluated in three forms: Raw (uncorrected), Comp (joint-compensated), and Supp (stage-adaptively stabilized), denoted as $T_{\text{Raw}}(t)$, $T_{\text{Comp}}(t)$, and $T_{\text{Supp}}(t)$, respectively. Because disturbance intensity varies substantially over the burning process, results are reported for the full process and also in a stage-wise manner for the preignition

combustion phase, flaming combustion phase, and smoldering combustion phase.

For unified expressions, the residual is defined as $e_k(t) = T_k(t) - T_{\text{ref}}(t)$, where $k \in \{\text{Raw, Comp, Supp}\}$. Trend consistency is quantified by the Pearson correlation coefficient $\rho_k = \text{corr}(T_k(t), T_{\text{ref}}(t))$, where a larger ρ_k indicates better ability of the TIR sequence to follow the reference trend. Accuracy is evaluated by $\text{RMSE}_k = \sqrt{\frac{1}{N} \sum_{t=1}^N e_k^2(t)}$, and the systematic drift is characterized by $\text{Bias}_k = \frac{1}{N} \sum_{t=1}^N e_k(t)$. To quantify short-term abnormal fluctuations under smoke and turbulence in a robust manner, we further compute a MAD-based scale estimator $\sigma_{\text{MAD},k} = 1.4826 \cdot \text{median}(|e_k(t) - \text{median}(e_k(t))|)$, which has the same unit as standard deviation but is less sensitive to outliers. These metrics are reported for both the full process and each combustion stage.

As an additional comparative experiment, a polynomial compensation model was introduced under the same thermocouple-aligned evaluation protocol. Following a conventional compensation-oriented strategy, the comparison model uses the recovered raw TIR temperature, observation distance, viewing angle, and smoke descriptor as inputs to estimate the temperature error. For the m th-order polynomial model

$$\Delta T_{\text{poly}}^{(m)}(t) = f_m(T_{\text{raw}}(t), d(t), \theta(t), \tau_s(t)), m = 1, 2, 3 \quad (21)$$

and the compensated temperature is written as

$$T_{\text{poly}}^{(m)}(t) = T_{\text{raw}}(t) + \Delta T_{\text{poly}}^{(m)}(t) \quad (22)$$

To make the comparison output consistent with the final sequence form used in this study, a fixed-parameter EWMA smoothing step was further applied:

$$T_{\text{cmp}}^{(m)}(t) = \alpha T_{\text{poly}}^{(m)}(t) + (1 - \alpha) T_{\text{cmp}}^{(m)}(t - 1) \quad (23)$$

where $f_m(\cdot)$ denotes an m th-order polynomial mapping, T_{raw} is the recovered raw TIR temperature, d is the observation distance, θ is the viewing angle, τ_s is the smoke descriptor, and α is the fixed EWMA smoothing coefficient. First-, second-, and third-order polynomial models were evaluated, and the best-performing one was used for comparison with the proposed method in the subsequent full-process analysis.

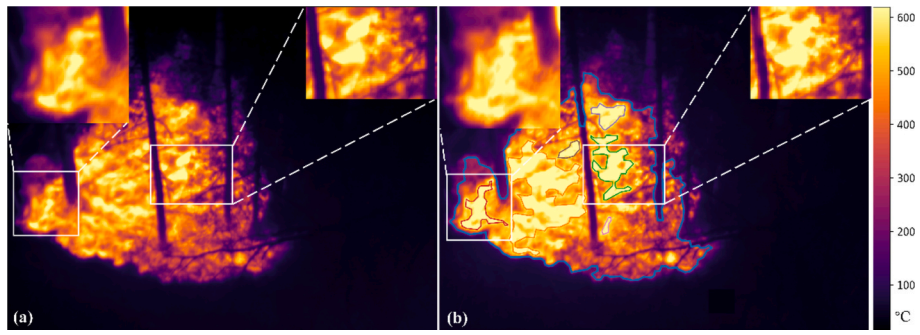


Fig. 7. Spatial comparison of temperature fields before and after joint compensation: (a) Before compensation; (b) after compensation, with details enlarged in the white box. In (b), the contours delineate different combustion stages.

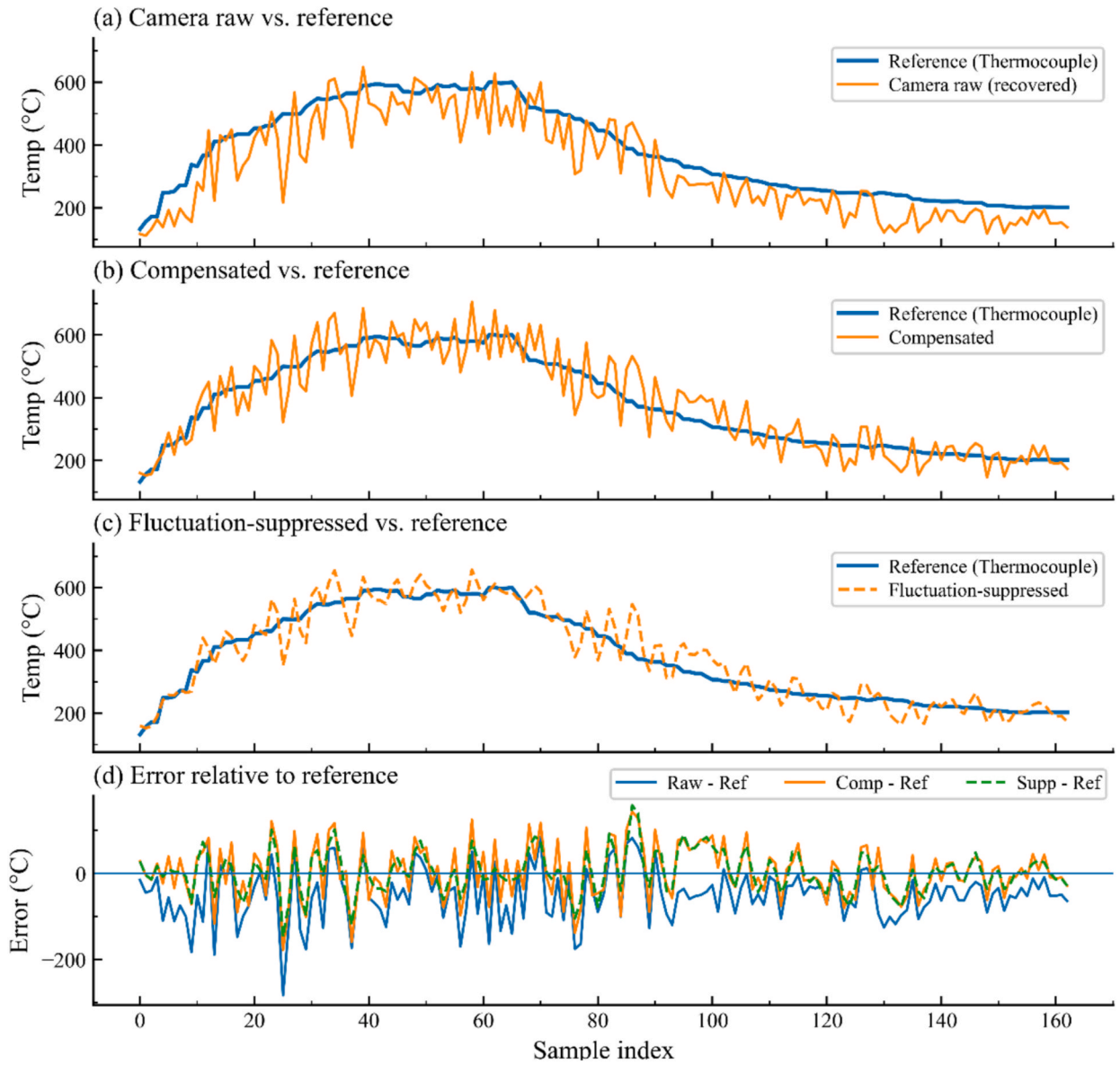


Fig. 8. Time-series comparison among raw, compensated, and stabilized sequences and the thermocouple reference: (a) recovered camera raw versus reference; (b) compensated versus reference; (c) stabilized versus reference; (d) temperature errors relative to the reference (raw-ref, comp-ref, and supp-ref) along the sample index.

Table 3

Full-process quantitative metrics for Raw, Comp and Supp.

Method	BMSE _k (°C)	Bias _k (°C)	ρ_k	$\sigma_{MAD,k}$ (°C)
Raw	78.5251	-49.1122	0.9191	59.0179
Comp	62.947	3.9384	0.9162	69.2908
Supp	47.8006	3.9333	0.9493	48.6185

3.3. Results and analysis

3.3.1. Spatial temperature-field effect after joint compensation

We first examine the effect of joint compensation from a spatial perspective. Fig. 7 compares representative temperature fields before and after compensation under the same color scale. After compensation, the high-temperature region becomes more coherent, and the temperature distribution better matches the expected stratified structure of an evolving fire front, comprising a hot core, a transition zone, and a cooler background. Importantly, the compensation mainly modifies the temperature magnitude while preserving the overall spatial pattern, providing a more reliable temperature field for subsequent interpretation and validation.

3.3.2. Time-series comparison and full-process quantitative results

Fig. 8 shows the continuous time-series comparison among Raw, Comp, and Supp sequences against the thermocouple reference. Here, the Raw, Comp, and Supp sequences are extracted at the back-projected thermocouple location on the TIR image and are computed as the spatial average over a 3×3 neighborhood centered at the projected pixel. Compared with Raw, the Comp sequence is shifted toward the correct magnitude level, indicating that the proposed joint compensation effectively reduces systematic drift introduced by distance, viewing geometry, and smoke attenuation. Meanwhile, short-term burst-like disturbances may still appear in Comp due to transient occlusion and rapidly varying observation conditions. After applying the stage-adaptive stabilization, these short-term abnormalities are substantially suppressed while the main combustion trend is preserved, leading to a smoother and more interpretable sequence that better follows the reference trend.

The full-process quantitative metrics are summarized in Table 3. Overall, joint compensation contributes primarily to bias reduction,

whereas the stage-adaptive stabilization further reduces the robust fluctuation scale and improves trend consistency without introducing additional systematic drift.

3.3.3. Scatter consistency over the full process

To provide an intuitive, statistical-domain view of tracking behavior, Fig. 9 presents the scatter consistency between the TIR-derived sequence and the thermocouple reference over the full process. Compared with Raw, Comp results show reduced systematic deviation; after stabilization, the scatter points become more concentrated around the identity line, consistent with the improvement in trend consistency and the reduction of abnormal fluctuations.

3.3.4. Stage-wise behavior across combustion phases

Given the strong temporal non-stationarity of wildfire burning, stage-wise evaluation is further performed for the preignition combustion phase, flaming combustion phase, and smoldering combustion phase. The stage-wise quantitative results are summarized in Table 4, and the corresponding scatter comparisons are shown in Fig. 10. The stage-wise analysis generally agrees with the full-process observations: joint compensation consistently reduces systematic bias across phases, while stage-adaptive stabilization yields a more pronounced reduction of short-term volatility and improved trend-following performance, particularly in the flaming combustion phase where disturbances are typically the strongest.

3.3.5. Comparison with a polynomial compensation model

To further evaluate whether the proposed framework provides advantages over a conventional compensation-oriented method, the polynomial compensation model described in Section 3.2 was compared with the final proposed output. First-, second-, and third-order polynomial models were all evaluated under the current dataset and protocol. Table 5 summarizes the full-process quantitative results of these comparison models together with the proposed method, and Fig. 11 presents the sequence-level comparison and temperature error comparison using the best-performing polynomial model.

As shown in Table 5, all polynomial compensation models improve the raw temperature sequence to different extents, indicating that a conventional compensation-oriented strategy can partially recover the wildfire temperature trend. Under the current dataset and evaluation

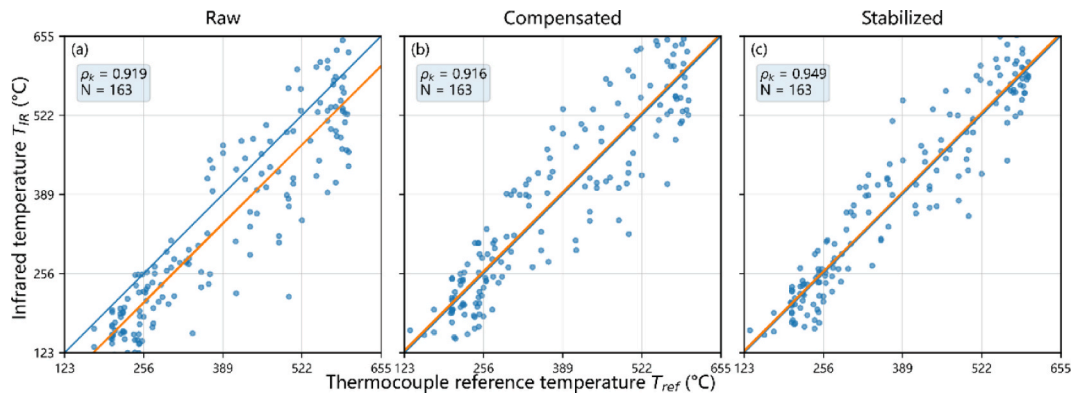


Fig. 9. Scatter consistency between TIR-derived sequences and the thermocouple reference: The correlation coefficient (ρ_k) is reported in each panel.

Table 4

Stage-wise quantitative metrics for preignition, flaming and smoldering combustion.

Stage Method	Preignition combustion			Flaming combustion			Smoldering combustion		
	Raw	Comp	Supp	Raw	Comp	Supp	Raw	Comp	Supp
BMSE _k	87.06	57.43	38.93	92.48	75.59	53.19	61.68	52.52	45.68
Bias _k	-56.08	3.28	3.30	-50.37	0.30	0.06	-45.93	7.08	7.25
$\sigma_{MAD,k}$	69.24	52.55	29.88	81.44	87.64	54.96	34.60	54.04	48.10

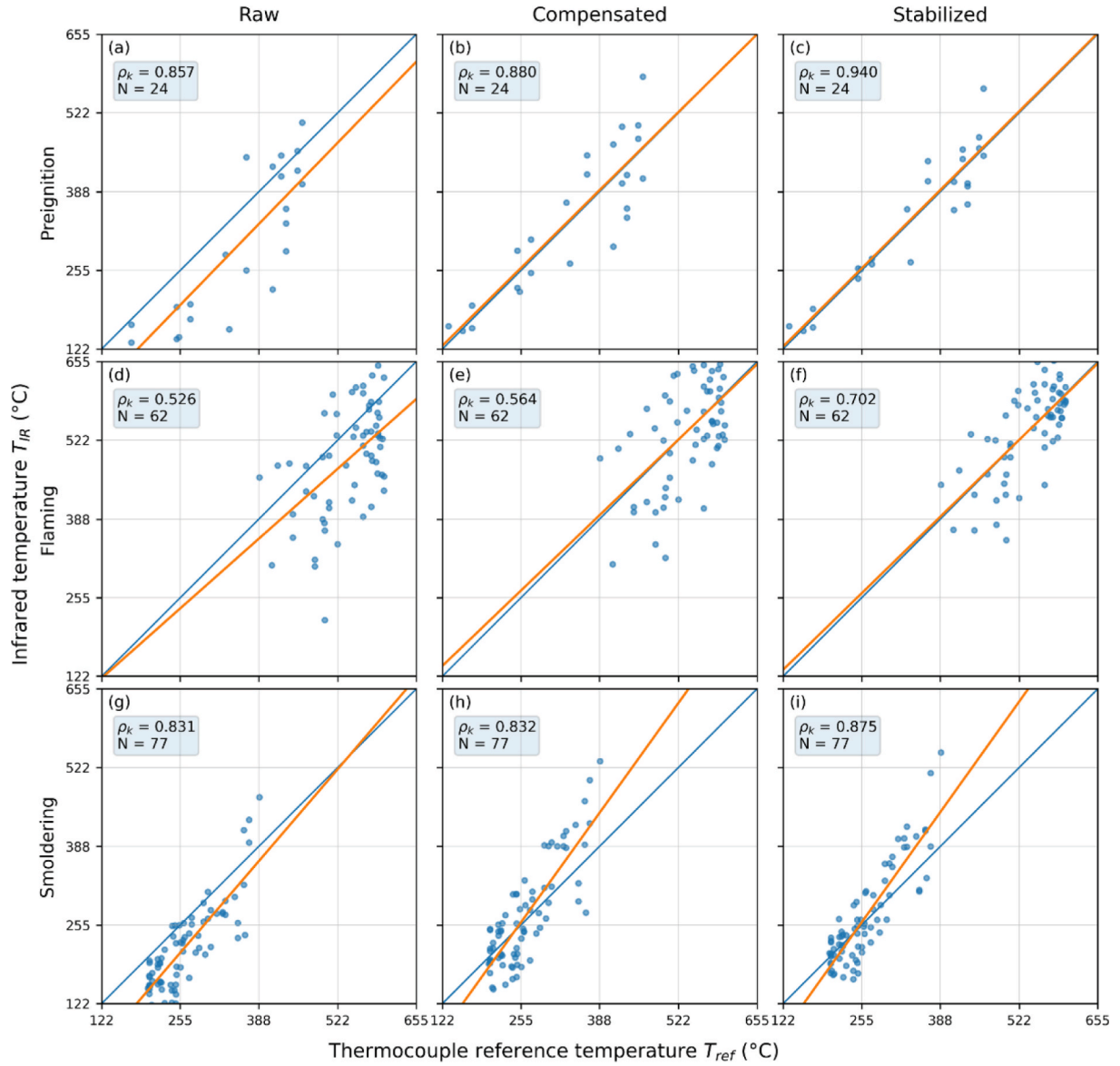


Fig. 10. Stage-wise scatter consistency for preheating, flaming, and smoldering combustion phases: Each panel reports ρ_k and the number of samples N .

Table 5

Full-process quantitative comparison between the polynomial compensation models and the proposed method.

Method	BMSE _k (°C)	Bias _k (°C)	ρ_k	$\sigma_{MAD,k}$ (°C)
1st-order polynomial compensation model	50.42	8.75	0.9257	51.67
2nd-order polynomial compensation model	55.16	12.77	0.9190	62.30
3rd-order polynomial compensation model	58.95	14.41	0.9131	69.58
Proposed method	47.80	3.93	0.9493	48.62

protocol, the first-order polynomial model achieves the best overall performance among the three comparison models, and is therefore used for the sequence-level comparison in Fig. 11. Nevertheless, the proposed method still yields better overall agreement with the thermocouple reference, with lower RMSE, lower bias, higher correlation, and a smaller robust fluctuation scale. In particular, the bias is further reduced from 8.75 °C to 3.93 °C, showing that the proposed framework more effectively suppresses the remaining systematic deviation while

maintaining stable sequence tracking.

3.3.6. Additional validation under different real-fire conditions

To assess whether the proposed framework remains effective beyond a single fire case, an additional validation experiment was conducted under different real-fire scenario. The newly introduced scenario differs from the first one in several representative aspects, including field environment, dominant combustion behavior, observation conditions, wind level, and fuel composition. Specifically, while the first scenario was acquired in an understory fire with a longer flaming-dominant period, the second scenario was acquired in a grassland fire under stronger wind conditions, with a longer char-dominated combustion stage. By applying the same thermocouple-aligned evaluation protocol and the same Raw-Comp-Supp processing framework to this second scenario, we aimed to examine whether the main improvement pattern of the method remains valid under different real-fire scenario.

As shown in Fig. 12 and Table 6, the proposed method remains effective under different real-fire scenario. The Raw sequence exhibits a large negative bias relative to the thermocouple reference, while the Comp sequence pulls the overall bias back toward zero, reducing it from −52.69 °C to 7.40 °C. The Supp sequence further improves the overall

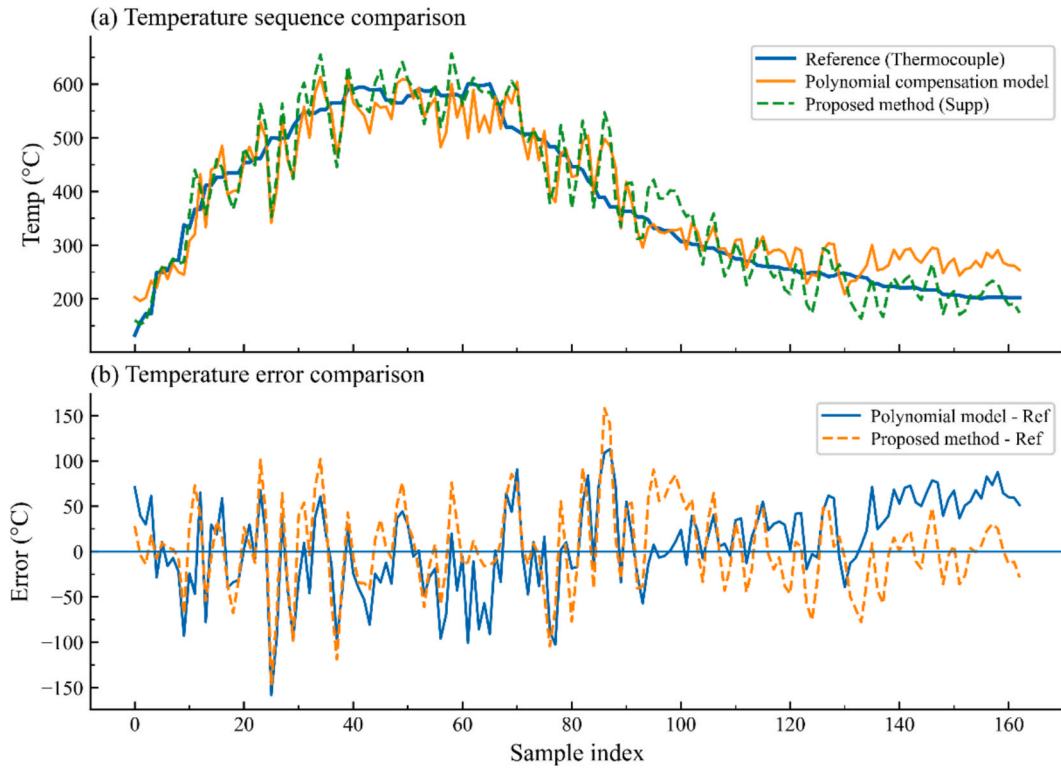


Fig. 11. Comparison between the polynomial compensation model and the proposed method: (a) temperature sequence comparison against the thermocouple reference; (b) temperature error comparison relative to the thermocouple reference.

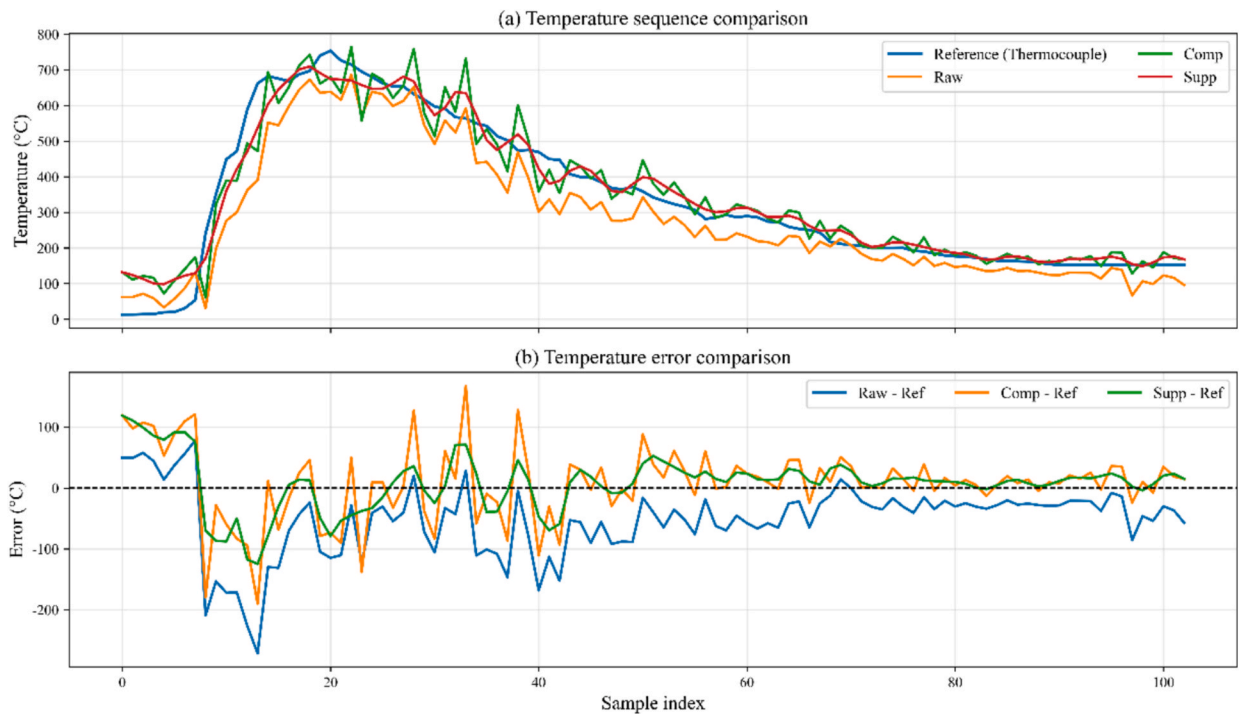


Fig. 12. Validation results in a second real-fire scenario: (a) temperature sequence comparison against the thermocouple reference; (b) temperature error comparison for Raw, Comp, and Supp.

sequence quality, with RMSE decreasing from 60.27 °C to 44.11 °C, correlation increasing from 0.9084 to 0.9313, and the robust fluctuation scale decreasing from 37.54 °C to 20.27 °C. These results show that the main improvement trend observed in the first scenario is reproduced in

the second scenario as well.

Table 6

Full-process quantitative results of Raw, Comp, and Supp in the second real-fire scenario.

Method	BMSE _k (°C)	Bias _k (°C)	ρ_k	$\sigma_{\text{MAD},k}$ (°C)
Raw	79.21	-52.69	0.9124	35.30
Comp	60.27	7.40	0.9084	37.54
Supp	44.11	7.43	0.9313	20.27

3.4. Discussion

The experimental results confirm that the proposed pipeline provides complementary benefits at different levels. Joint compensation mainly corrects the systematic drift induced by distance-dependent extinction, viewing-geometry effects, and smoke attenuation, thereby pulling the TIR magnitude toward the thermocouple reference (see Table 3 and Table 4). In contrast, stage-adaptive stabilization primarily improves reliability by suppressing short-term abnormal fluctuations while preserving the main combustion trend.

The newly added comparison with polynomial compensation models further clarifies the benefit of the proposed framework from the perspective of conventional compensation-oriented methods. Although all polynomial models improve the raw temperature sequence to different extents, the proposed method still achieves better overall agreement with the thermocouple reference. In particular, the first-order polynomial model gives the best performance among the comparison models, but the proposed method still yields lower RMSE, lower bias, higher correlation, and a smaller robust fluctuation scale. This indicates that the advantage of the proposed framework does not come only from an additional smoothing step, but from the coordinated effect of multi-factor joint compensation and combustion-stage-aware stabilization.

In real-fire observations, the influencing factors—particularly smoke transmittance and observation geometry—can vary abruptly due to transient occlusion, flame intermittency, and platform micro-perturbations. As a result, after removing the dominant bias component, the compensated sequence can exhibit more apparent high-frequency residuals or burst-like deviations. This explains why Comp does not necessarily yield the lowest volatility, even when its magnitude accuracy is improved. The stage-adaptive stabilization addresses this issue: the Hampel–MAD step robustly rejects impulsive deviations, and the EWMA step reduces remaining high-frequency disturbance. Importantly, the stabilization strength is controlled by the combustion-stage label, enabling a different balance between responsiveness and suppression across the preignition combustion phase, flaming combustion phase, and smoldering combustion phase.

The second real-fire validation helps explain the robustness of the proposed framework from a mechanism perspective. Although the two fire cases differ in field environment, dominant combustion behavior, observation conditions, wind level, and fuel composition, the same separation between compensation and stabilization is still observed. This suggests that the compensation module mainly addresses the large low-frequency drift introduced by multi-factor observation effects, whereas the stabilization module mainly acts on the residual short-term disturbances that remain after bias correction. Within the tested range, this role separation appears to be stable across different real-fire conditions rather than being specific to a single fire case.

Several limitations should still be noted. First, although the added second real-fire validation strengthens the support for applicability under different conditions, the present study does not claim full universality across all wildfire environments; the validated range is still limited to two real-fire scenarios. Second, the separable approximation of the joint transfer coefficient assumes that distance, viewing geometry, and smoke effects can be factorized, whereas residual coupling may still exist in practice and is absorbed into the fitted parameters. Third, smoke transmittance is inferred from visible imagery and mapped to thermal-

band attenuation through an effective exponent, which may vary with particle properties and illumination conditions. Fourth, a spatial-support mismatch remains between point-wise thermocouple measurements and line-of-sight radiative observations, although neighborhood averaging alleviates this issue. Under broader wildfire conditions, practical adjustment strategies may include recalibration of compensation coefficients, re-identification of the smoke-related mapping, and moderate retuning of stage-label thresholds or smoothing strength. Future work will therefore focus on broader cross-scenario validation, improved smoke characterization, and more explicit radiative-transfer modeling.

4. Conclusions

This paper presents a UAV-based infrared temperature correction framework for dynamic wildfire observations, integrating multi-modal sensing and physics-guided modeling. First, key influencing factors are quantified at the pixel level, including observation distance, viewing angle, and smoke transmittance. Second, a radiance-increment formulation is used to construct a multi-factor joint compensation model, where the equivalent transfer coefficient is decomposed into distance-, angle-, and smoke-related sub-functions and identified through calibration and smoke-involved data. Third, a combustion-stage-aware stabilization module is introduced to suppress short-term abnormal fluctuations via Hampel–MAD outlier suppression followed by stage-dependent EWMA smoothing.

Real-fire experiments show a clear progression from uncorrected to joint-compensated and then to stage-adaptively stabilized. Joint compensation mainly reduces the dominant systematic drift, while stabilization further improves sequence reliability by suppressing burst-like deviations and enhancing trend tracking. Additional comparison with polynomial compensation models shows that the proposed method provides better overall agreement with the thermocouple reference than conventional compensation-oriented models. Moreover, validation under different real-fire conditions confirms that the same two-stage improvement pattern is retained across two fire cases with clearly different field and combustion characteristics. Overall, the proposed framework provides an effective and practical solution for improving continuous UAV-based wildfire temperature monitoring and can serve as a reliable temperature basis for downstream fire-front analysis and combustion characterization.

CRediT authorship contribution statement

Yue Ma: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Siyu Liu:** Writing – review & editing, Validation, Investigation, Data curation. **Haiyu Zhang:** Writing – review & editing, Resources, Investigation, Data curation. **Dandan Li:** Writing – review & editing, Validation, Investigation. **Hongyang Zhao:** Visualization, Investigation, Data curation. **Boqiang Wang:** Resources, Investigation. **Xingdong Li:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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